Gokhale Education Society's

R. H. Sapat College of Engineering, Management Studies and Research,
Nashik - 422 005, (M.S.), INDIA

Computer Engineering

ATM Management System

By
Chaitanya Mahesh Deshpande
Roll No. : 33
Group No.9

Under the guidance of
Ms . Vrushali Nikam

ATM Java Program with MySQL Connectivity

Simulating ATM functionalities using Java and MySQL. Developed in VS Code. Provides secure and reliable banking operations.

Table of Contents

Problem Statement

Introduction

Objectives

Drawbacks of Existing System

Software Requirements

Scope of Project and Conclusion

Outline of the project's key areas. We will cover each section in detail. Follow along as we explore each topic.

Problem Statement

1. **Inefficient Processes**

   Manual banking leads to long wait times.

2. **Limited Accessibility**

   Lack of 24/7 availability for basic transactions.

3. **Security Risks**

   Traditional methods have security vulnerabilities.

4. **High Costs**

   Manual systems have high operational costs.

Addressing these issues can greatly enhance banking.

Introduction

1

ATM System

Automated Teller Machine overview.

2

Java Backend

Java for backend logic.

3

MySQL Storage

MySQL for secure data storage.

4

VS Code IDE

VS Code as the IDE.

ATMs are vital in modern banking, with over 400,000 in the US.

Objectives

User-Friendly Interface

Develop an easy-to-use ATM interface.

Secure Processing

Implement secure transactions with encryption.

MySQL Connection

Connect to MySQL for account management.

Automation can reduce transaction costs by 40%.

Drawbacks of Existing Systems

High Costs

Maintenance averages $1000/year per machine.

Limited Functionality

Outdated interfaces restrict capabilities.

Fraud Risk

Outdated security measures increase fraud.

Addressing these issues is critical for improvement.

Software Requirements & Project View

1. **Java JDK**
   JDK 8 or higher is required.

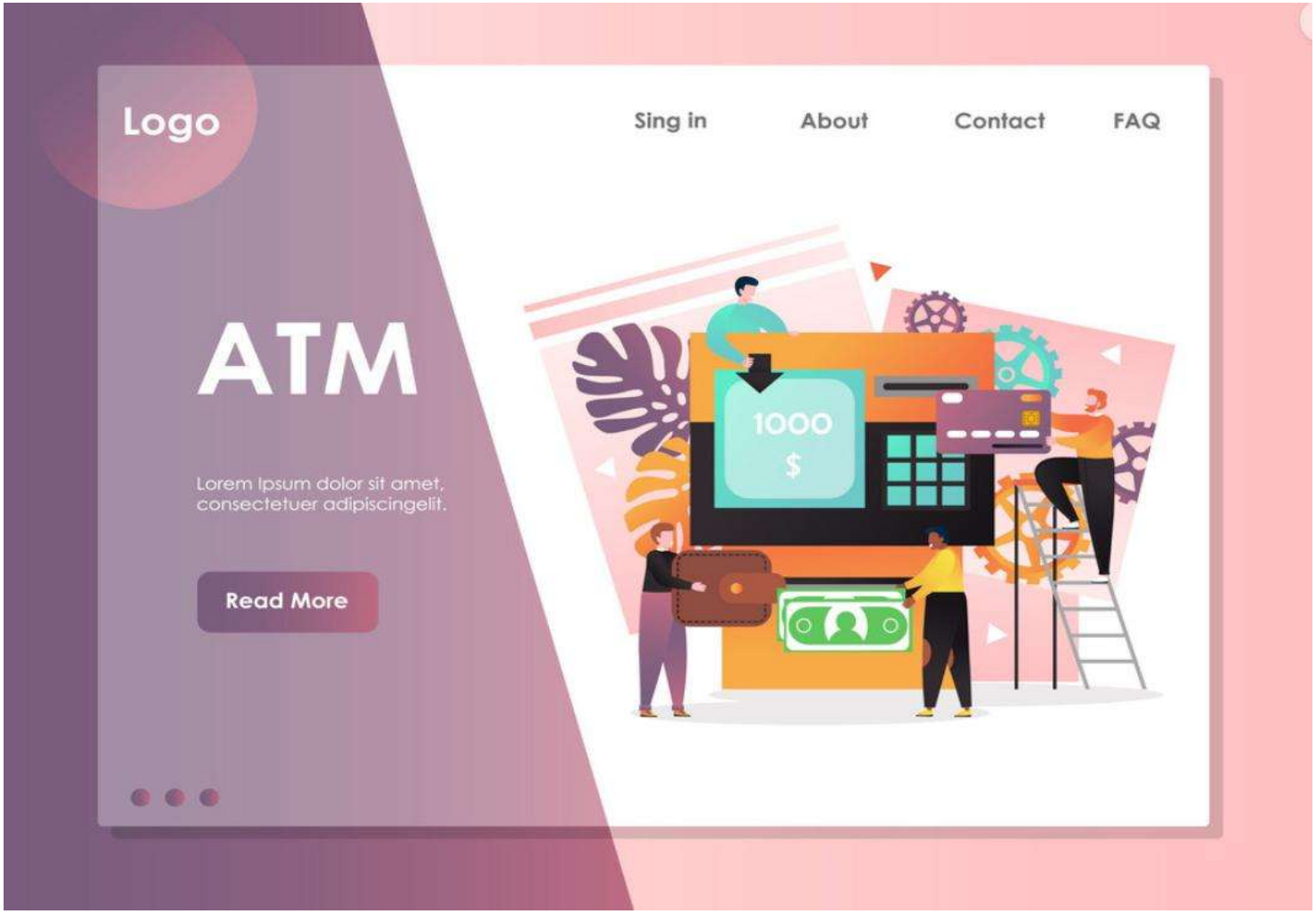
2. **MySQL Server**
   MySQL Server 5.7 or higher is needed.

3. **VS Code**
   VS Code with Java and MySQL extensions.

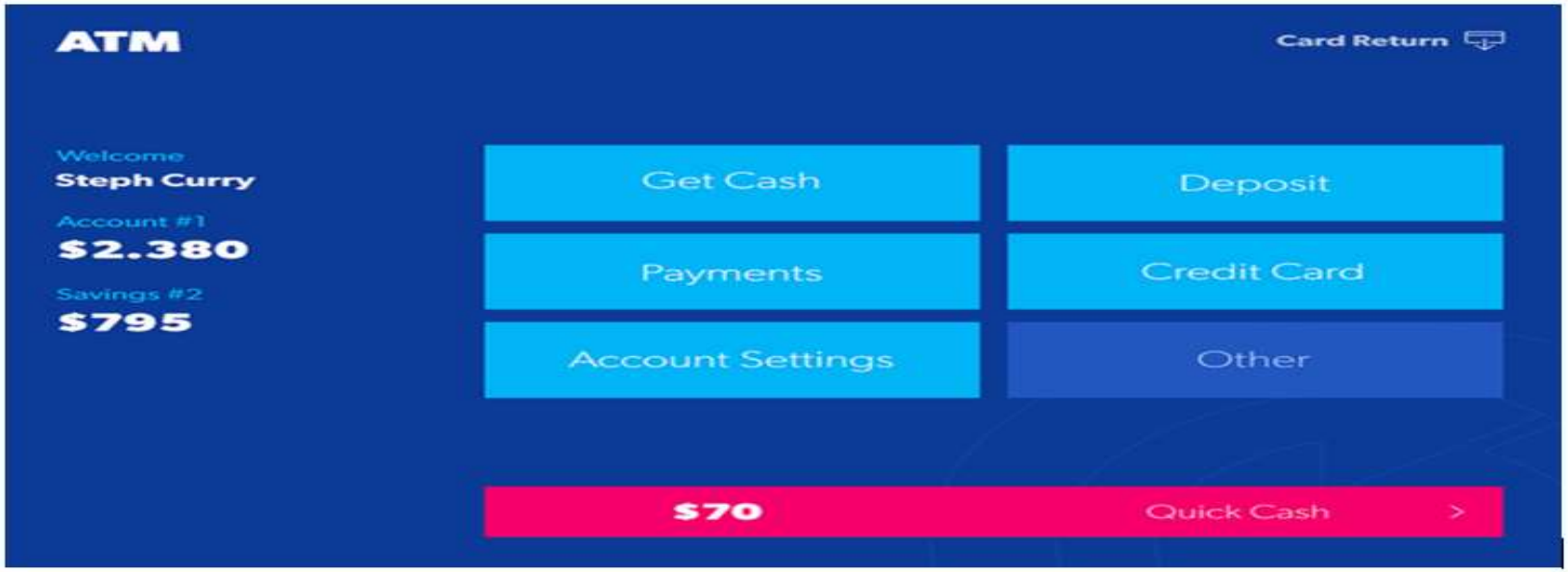
4. **JDBC Driver**
   MySQL Connector/J JDBC driver.

Screenshots show the ATM interface and database schema.

Home Page :


Logo
Sing in
About
Contact
FAQ
ATM
Lorem Ipsum dolor sit amet,
consectetuer adipiscingelit.
Read More
1000
$

About Us –



Automated teller machine

An **automated teller machine** (ATM) is an electronic banking outlet that allows customers to complete basic transactions without the aid of a branch representative or teller. Anyone with a credit card or debit card can access cash at most ATMs, either in the U.S. or other countries.

Login Page :

AUTOMATED TELLER MACHINE
WELCOME TO ATM
Card no. :
PIN :
SIGN IN
CLEAR
SIGN UP

Java Program :

```java
import java.util.Scanner;

class Bank {
    private double balance;

    public Bank(double initial) {
        this.balance = initial;
    }

    public void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Deposited: " + amount);
        } else {
            System.out.println("Invalid amount!");
        }
    }

    public void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient balance or invalid amount!");
        }
    }

    public void check() {
        System.out.println("Balance: " + balance);
    }
}

class ATM {
    private Bank bank;
    private Scanner scan;

    public ATM(Bank bank) {
```

```java
class ATM {

    public ATM(Bank bank) {
        this.bank = bank;
        this.scan = new Scanner(System.in);
    }

    public void start() {
        while (true) {
            System.out.println("\n1. Withdraw\n2. Deposit\n3. Check Balance\n4. Exit");
            System.out.print("choose: ");
            int choice = scan.nextInt();

            switch (choice) {
                case 1:
                    System.out.print("Enter amount: ");
                    bank.withdraw(scan.nextDouble());
                    break;
                case 2:
                    System.out.print("Enter amount: ");
                    bank.deposit(scan.nextDouble());
                    break;
                case 3:
                    bank.check();
                    break;
                case 4:
                    System.out.println("Exiting...");
                    return;
                default:
                    System.out.println("Invalid choice!");
            }
        }
    }
}

public class ATM_interface {
    Run | Debug
```

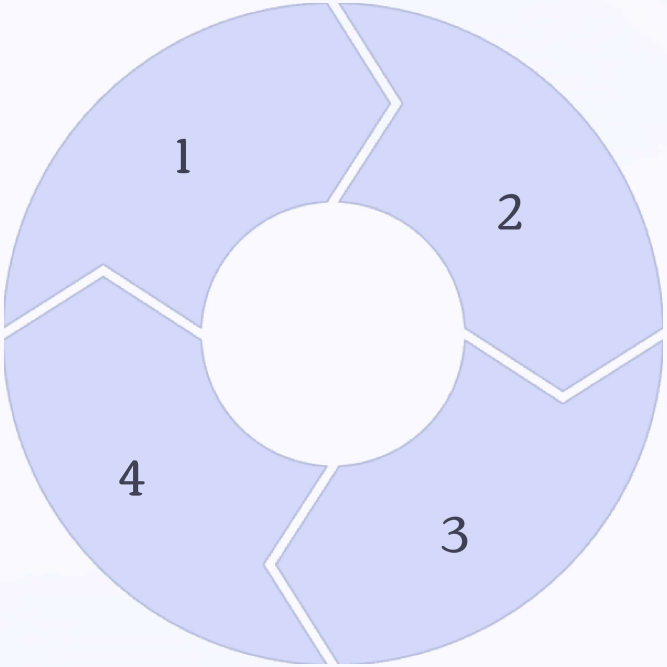
Scope of Project & Conclusion

Authentication

User authentication.

Transactions

Balance inquiry.



Security

Fraud detection.

History

Transaction history.

Enhances banking efficiency and reduces operational costs. Secure transactions via MySQL connectivity provide reliability.

C.ID: 98fd6cd

CodSoft

CERTIFICATE

OF COMPLETION

PROUDLY PRESENTED TO

Chaitanya Mahesh Deshpande

has successfully completed 4 weeks of a virtual internship program in

Java Programming

with wonderful remarks at **CODSOFT** from **10/01/2025 to 10/02/2025.**

We were truly amazed by his/her showcased skills and invaluable contributions to the tasks and projects throughout the internship.

Founder

CERTIFIED ISO 9001-2015 COMPANY

MSME MICRO, SMALL & MEDIUM ENTERPRISES

contact@codsoft.in

www.codsoft.in

Date: 13/02/2025

THANK
YOU